



DR. EMILY HATT

63 Russell Road, Birmingham, B13 8RB

📞 07342650441 ✉️ ehatt98@gmail.com



Postdoctoral researcher in the field of Asteroseismology. Scientific expertise includes generating an automated pipeline to detect solar-like oscillations, using asteroseismology to measure core rotation and magnetism, employing hierarchical Bayesian modelling and emulating grids of stellar models using a neural network. Proficient software developer, especially skilled in Python.

SCIENTIFIC POSITIONS

Postdoctoral Researcher, ISTA

Prof. Lisa Bugnet

June 2026 - present

Research Fellow, University of Birmingham

Prof. William Chaplin, Prof. Guy Davies

Jul 2024 - May 2026

PhD Astrophysics, University of Birmingham

Supervised by Prof. William Chaplin

Asteroseismic Inference on a Population Scale: From Detection to Internal Properties

Sep 2020 - Jun 2024

EDUCATION

University of Birmingham, MSci Physics + Astrophysics

Result: 1st Class Honours

Sep 2016 - Jul 2020

The Sixth Form College Farnborough

Sep 2014 - Jul 2016

KEY PUBLICATIONS

See QR code at the top of this page for a full list of publications.

Hatt, E., J. M. J. Ong, M. B. Nielsen, *et al.*, “Asteroseismic signatures of core magnetism and rotation in hundreds of low-luminosity red giants,” vol. 534, no. 2, pp. 1060–1076, Oct. 2024. DOI: 10.1093/mnras/stae2053. arXiv: 2409.01157 [astro-ph.SR]

Hatt, E., M. B. Nielsen, W. J. Chaplin, *et al.*, “Catalogue of solar-like oscillators observed by TESS in 120-s and 20-s cadence,” vol. 669, A67, A67, Jan. 2023. DOI: 10.1051/0004-6361/202244579. arXiv: 2210.09109 [astro-ph.SR]

M. B. Nielsen, J. M. J. Ong, **Hatt, E. J.**, *et al.*, “Asteroseismology with PBJam 2.0: Measuring Dipole Mode Frequencies in Coupling Regimes from Main-sequence to Low-luminosity Red Giant Stars,” vol. 169, no. 6, 322, p. 322, Jun. 2025. DOI: 10.3847/1538-3881/adcb37. arXiv: 2506.20382 [astro-ph.SR]

J. Ong, M. Nielsen, **Hatt, E.**, *et al.*, “Reggae: A Parametric Tuner for PBJam, and a Visualization Tool for Red Giant Oscillation Spectra,” *The Journal of Open Source Software*, vol. 9, no. 99, 6588,

p. 6588, Jul. 2024. DOI: 10.21105/joss.06588. arXiv: 2407.09956 [astro-ph.IM]

M. B. Nielsen, **Hatt, E.**, W. J. Chaplin, *et al.*, “A probabilistic method for detecting solar-like oscillations using meaningful prior information. Application to TESS 2-minute photometry,” vol. 663, A51, A51, Jul. 2022. DOI: 10.1051/0004-6361/202243064. arXiv: 2203.09404 [astro-ph.SR]

W. H. Ball, A. H. M. J. Triaud, **Hatt, E.**, *et al.*, “Projected spin-orbit alignments from Kepler asteroseismology and Gaia astrometry,” vol. 521, no. 1, pp. L1–L4, Jun. 2023. DOI: 10.1093/mnras1/slad012. arXiv: 2301.10308 [astro-ph.SR]

M. B. Nielsen, G. R. Davies, incl **Hatt, E.**, *et al.*, “Simplifying asteroseismic analysis of solar-like oscillators. An application of principal component analysis for dimensionality reduction,” vol. 676, A117, A117, Aug. 2023. DOI: 10.1051/0004-6361/202346086. arXiv: 2306.13577 [astro-ph.SR]

M. J. Goupil, C. Catala, incl **Hatt, E.**, *et al.*, “Predicted asteroseismic detection yield for solar-like oscillating stars with PLATO,” vol. 683, A78, A78, Mar. 2024. DOI: 10.1051/0004-6361/202348111. arXiv: 2401.07984 [astro-ph.SR]

TALKS

Cool Stars 23 (conference)

2026, invited

Recent Observational Advances insights into Post Main-Sequence Rotation and Magnetism Review talk for splinter session [20 mins]

CFSA Series, Warwick

2026, invited

Seeing into Stars with Asteroseismology and Advanced Statistics - The Missing Angular Momentum Transport Problem.

ISTA Astronomy Seminar Series

2025, invited

Seeing into Stars with Asteroseismology and Advanced Statistics

Talk given to the Astronomy department at the Institute of Science and Technology, Austria [1 hour]

Asteroseismology for Exoplanet Scientists

2025, internal

Asteroseismology: Fundamentals and Applications

Talk given to University of Birmingham research group. [1 hour]

Astronomy on Tap

2025, invited

Dancing with the Stars: Stars, Sound and Me.

Public outreach talk about stellar physics and asteroseismology.

Tea, Talk and Telescope

2024, invited

Stars, Sound and Me

Seminar given to University of Birmingham Astronomical Society. [1 hour]

Tutorial on the use of Canva for presentations

2024, internal

Giving memorable presentations; from content to visuals with Canva

Session delivered to members of University of Birmingham research group presenting tips and tricks for giving memorable presentations, and how to use the design tool *Canva* to make interesting slides [1 hour]

Asteroseismic Inference on a Population Scale

2024, internal

Overview of my PhD

Talk given to University of Birmingham research group. [1 hour]

Asteroseismology Tutorial

2023, internal

Introduction to Asteroseismic Data Analysis

Session delivered to members of University of Birmingham research group dedicated to introducing asteroseismology and walking attendees through some basic analysis tools, including **pbjam** and **Lightkurve**. [1 hour]

TASC7/KASC14 (conference)

2023

An Ensemble View of Magnetic Imprints on Mixed Modes at the Base of the RGB

TESS mission Update Meeting (conference)

2023, invited

A Catalogue of Solar-Like Oscillators in Short Cadence

SASP EDI Event

2022, organised

Space is Black: A Celebration of Black Astrophysicists

Event marking the unveiling of a set of posters I lead in designing to promote inclusivity. Gave a talk about the status of inclusivity in Astronomy.

Asterochronometry Seminar Series

2022, invited

A probabilistic method for detecting solar-like oscillations and results in TESS data

Virtual seminar for the Asterochronometry group at the University of Bologna. [30 mins]

TASC6/KASC13 (conference)

2022

Asteroseismology with TESS Short Cadence Data: Finding thousands of new solar-like oscillators using a probabilistic detection algorithm

Good Vibrations Seminar Series

2021, invited

Asteroseismology with TESS; Bridging the Gap Between The Red Giant Branch and the Main Sequence

Virtual seminar hosted by Université PSL. Presentation viewable online. [45 mins]

National Astronomy Meeting (conference)

2021

TESS Asteroseismology: New Detections from Main-sequence to Subgiant Stars

TEACHING

Summer Studentship: Physics and Art

Sept 2025

Supervised a summer studentship project combining art and astrophysics.

Undergraduate 1st Year Astrolab

2025

Guided students through astrophysical laboratory projects and contributed to developing the laboratory manual.

Undergraduate 3rd Year Group Studies

2023

Supervised a group project concerning the asteroseismic potential of PLATO. Headed the modelling team, consisting of 9 students.

Undergraduate 1st Year Mathematics Problem Classes

2021 - 2023

Guided students through mathematics problems. Involved both one-to-one support and demonstrating solutions to the class.

Undergraduate 2nd Year Computing Lab

2021 - 2023

Supported students in learning how to code in Python. Introduced the Jupyter Notebook in the process.

OUTREACH

Offer Holder Visit Days

2022 - 2023

Ran 45 minute Astrophysics workshops held for students with offers for undergraduate study at the University of Birmingham. Attended on average one per semester.

Workshops for Home Educated Students*2022 - 2023*

Aided in running educational workshops on campus for home educated students.

Lead Organiser of E&D Committee*2021 - 2023*

Headed organisation of the Birmingham Sun, Stars and Exoplanets Equality and Diversity Committee. Lead project producing posters highlighting astronomers from underrepresented groups. Organised an unveiling event, with an invited speaker. Aided in developing several school workshops aimed at increasing inclusion in astronomy, which were delivered at a number of events.